

# **LAB 1: USE-CASE FLOW SPECIFICATION**

## **Problem Statement #61 – Sustainability & Green Tech**

### **Household Carbon Footprint & Target Tracker**

**Name:** MUKESH SHEREGAR

**SRN:** PES1UG25CS828

**Section:** I

#### **Use Case: UC-02 – Calculate Monthly CO<sub>2</sub>e**

Primary Actor: Resident

Supporting Actor: Emission Factor Service / Database

Goal: Calculate the household's total monthly CO<sub>2</sub>e emissions using recorded electricity, gas and transport data.

#### **Preconditions**

- Resident is using the Household Carbon Footprint & Target Tracker.
- A month is selected for the calculation.
- Required monthly consumption data is available or can be entered.

#### **Postconditions**

- Monthly CO<sub>2</sub>e is calculated and displayed.
- The calculated result is stored for future tracking.
- The result can be used for historical comparison and milestone tracking.

#### **Main Success Scenario**

1. Resident selects the month for which the carbon footprint has to be calculated.
2. System displays the electricity, gas and transport consumption values.
3. Resident enters or confirms the monthly consumption values.
4. System validates the entered values.
5. System obtains the required conversion factors.
6. System calculates the CO<sub>2</sub>e contribution of each category.
7. System calculates the total monthly CO<sub>2</sub>e.
8. System displays the total CO<sub>2</sub>e and category-wise contributions.
9. System stores the calculated result.
10. Use case ends successfully.

#### **Alternate Flow – Invalid Consumption Data**

- 4a. System detects invalid or missing consumption data.
- 4a1. System displays an error message for the affected field.
- 4a2. Resident corrects the value.
- 4a3. System validates the corrected value.
- 4a4. If valid, the system continues from Step 5.
- 4a5. If valid data is not provided, the calculation is cancelled and no result is stored.